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The role of renewable energy production on greenhouse gas emission reduction in international petroleum companies under energy transition policy

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Abstract
This study explores the impact of renewable energy production on GHG emissions under energy transition in a panel of 27 international petroleum companies over the period of 2015–2023. For this purpose, we utilize the dynamic generalized method of momentum in the framework of Porter hypothesis. We find strong evidence that renewable energy production positively affects GHG emissions reduction in the short and long run. The outcomes also disclosed that the impact of research and development on GHG emissions reduction is only valid in the short run. Interestingly, both renewable energy production and research and development has an inverted U-shaped pattern in the relationship we identified from their side to GHG emissions. Furthermore, we find that research and development play a fundamental role in mediating the relationship between REP and GHG emissions. Finally, the effect of REP on GHG emissions is found to be more preannounced in EU international petroleum companies relatively to non-EU international petroleum companies. The study provides valuable information for policymakers and petroleum companies' managers.

Keywords Renewable energy production, GHG emissions, Porter hypothesis and GMM approach

1 Introduction

Global warming has been increasingly affecting each aspect of our live [27, 65]. Undoubtedly, greenhouse gas (GHG) emissions from economic activities that rely on the composition of fossil fuels for energy have been the prime driver of global warming [3, 68]. For this reason, the 2015 UN Climate Change Conference (COP21) in Paris established the global of net zero emission by 2050. To facilitate achieving this goal, the National Determined Contributions (NDCs) were introduced, outlining country-specific commandments to reduce national emissions. Subsequently, climate initiatives and strategies have been adopted within economic sectors around the world. Despite of these national and global strategies, GHG emissions have continued to rise alarmingly. In 2023, the highest

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level of GHG emission in human history was recorded with about 54 billion Tonnes of Carbon Dioxide Equivalents (tCO_2e). In this context, petroleum industry, which provide over 80% of the world's energy needs (ÓhAiseadha et al. 2020), is considered one of the largest polluting industry as it is responsible for 37% of global GHG emissions due to high oil and gas extraction and production [25]. For example, In 2023, About 100 million barrels of oil equivalent and 4,159 billion cubic metre of natural gas per day were extracted by petroleum companies [26]. Of course, this extraction is accompanied by a concomitant pollution in the forum of GHG emission, particularly scope 1 and 2. More specifically, scope 1 includes the direct GHG emissions from sources that are owned or controlled by petroleum companies. These are emissions that are readily-identifiable and directly attributable to a company's operations such as flaring, fugitive, leakage and the combustion needed on drilling. While Scope 2 is the indirect GHG emissions associated with the generation of purchased energy consumed by petroleum companies. For example, over the period of our study, the International Petroleum Companies (IPCs) in our sample emitted approximately 6897 million tCO_2e from Scope 1 and 2 sources, See Fig. 1 (Annual Sustainability reports). Thus, it was reported that only 25 IPCs are behind more than 50% of global industrial GHG emissions since 1998 to 2017 [13]. The latter report also stated that if these companies continue to extract fossil fuels at the same rate over the next 28 years, then global average temperatures would increase to 4 °C by the end of the century. Based on these alarming statements, in its updated report, the IEA [26] stated that there should be no new investments or extension of existing investments in petroleum to achieve net-zero emissions by the mid of this century. This underscores the crucial role that IPCs can play in achieving positive environmental outcomes. Their actions across production, operations, and innovation, particularly in developing renewable energy, are essential to underpin the implementation of NDCs, and thereby addressing the pressing global challenge [21, 54].

Recently, the increase awareness from the stakeholders such as civil society, managers and investors vis-a-vis the adverse impact of GHG emissions on environment [10], coupled with the tightening of regulatory frameworks such as the PA [22], makes maney IPCs recognize the importance of environmental protection and a low-carbon emission transition [42]. As a consequence, soem IPCs had adopted net zero emission goal (see Fig. 2) and started immensely diversifying their business models by deploying renewable energy in order to meet the timeline of net zero emission by 2050. For example, at

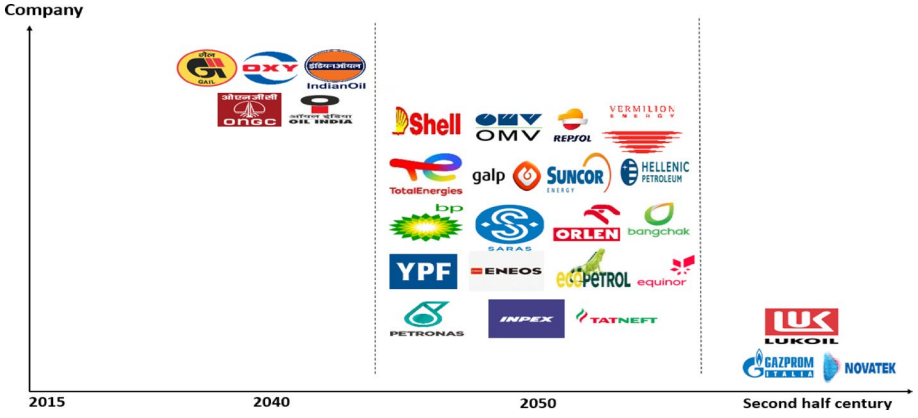


Fig. 1 Net zero emission goal of the IPCs

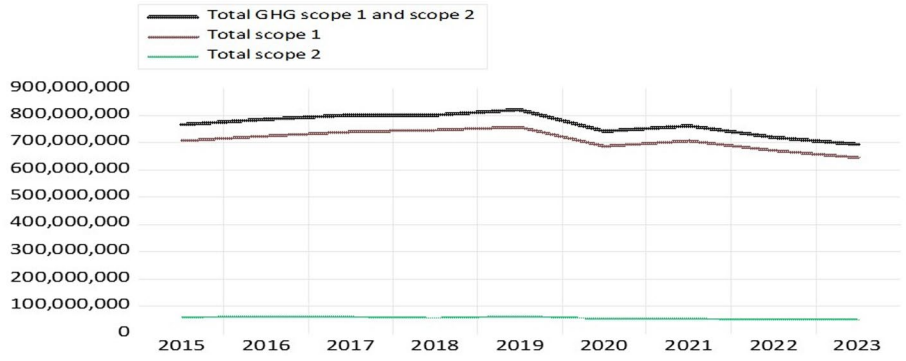


Fig. 2 Total scope 1 and 2 GHG emission of the chosen sample

the beginning of 2016, Equinor launched a new venture capital of $ 200 million to invest in renewable energy, in particularly wind power (Statoil Sustainability Report, 2016). One year later, Equinor has a Dudgeon and Hywind in Scotland in addition of having a start-up of Arkona and Apodi. As a result, the self-generation of Renewable Energy Production (REP) was 830 and 1,251 GWh in 2017 and 2018 respectively (Annual sustainability report, 2018). Statoil provides renewable electricity from offshore wind farms in German and UK to more than one million European homes. Likewise, the Canadian Suncor's RE business includes investments in five operating wind facilities and ethanol plant production with the capacity to generate 187 MW of electricity in 2016. In the same vein, Hellenic Energy showed that the initial production of renewable energy from the operation of photovoltaic stations exceeded 27 GWh in 2015, while the wind energy produced in Messinia exceeded 89 GWh, resulting in the avoidance of over 115, 000 tons of CO_2 emissions in 2017 (Annual sustainability reports 2015 and 2017). In 2019, Hellenic Energy also builds the photovoltaic park in Kozani, one of the largest RE projects in the EU, with a capacity of 350 GWh of energy production per year, capable of providing clean zero emission energy to 750,000 household. This resulted in avoiding 300,000 tons of CO_2 emissions. Also, Oil India Limited invested approximately 135 million $ in RE projects in 2015 aiming to become a net zero emission company by 2040 (Annual sustainability report, 2015).

Research and Development (R&D) is another crucial element of IPCs to achieve PA goal because it involves the use of new technology capable of not only controlling GHG emissions but also boosting REP, thereby accelerating the energy transition path [5, 19, 29, 41, 45]. For instance, Eni S.p.A' R&D expenditure amounted to 197.2 million € in 2018 which resulted in 43 patents, out of this 13 was related to renewable energy sources (Annual report, 2018). OMV Aktiengesellschaft R&D expenses were 61 million € in 2020, part of it was attributed to hydrogen, advanced fuel, co-processing, CCUS and energy storage. IndianOil Corporation in 2016, spent about 9.34 million $ on R&D resulted in 105 patents in cutting edge technology of low carbon technology with specific concentration on renewables such as solar, bio-energy and thermal and hydrogen (Sustainability report, 2016). Oil and Natural Gas Corporation (ONGC) spent about 74 million $ to upscale the digitalization across the business process and has filed 14 patents through its seed fund to 19 start-ups in 2019 (annual sustainability report, 2019). R&D include also energy efficiency projects which result in emerging of new environment-friendly technologies [43]. For instant, to achieve the goal of becoming a net-zero

emission, Repsol spent about 84 million € in projects includes carbon capture, utilization and storage (CCUS), e-fuel, green hydrogen and natural carbon sink, which led to about 20 patents at the end of 2018 (Repsol integrated report, 2018). Repsol also announced that it would invest 400 million euro from 2020 to 2025 in energy efficiency alone to avoid 800,000 metric tons of CO_2 annually (sustainability report, 2020).

To this end, our prime objective in this study is to explore the impact of REP on GHG emissions in petroleum industry. Exploring the impact of REP on GHG emission reductions to determine this impact are clearly missing in the related empirical research [2, 18, 38]. To prove, the majority of studies conducted on this subject are survey studies on one hand [1, 8, 11, 18, 41, 42, 44, 51, 59, 61]. On the other hand, empirical studies in this subject were conducted at the macroeconomic level (Saleh et al. 2024). This raises the question of whether REP reduced GHG emissions within IPCs or not. Our unique, hand-collected dataset on IPCs enable us to fill this gap.

Despite the fact that humanity is around 25 years away from achieving net-zero emissions, there remains a significant lack of empirical evidence to guide IPCs on how to effectively mitigate GHG emissions and pursue the goal of net zero emission [1, 8, 59]. Therefore, this study is motivated by the urgent need to develop and validate sustainable practice that can guid IPCs not only to mitigate the adverse effects caused by their GHG emission but also to achieve net zero emission by the mid of this century. We made multiple contributions to the related literature in several aspects. First and foremost, this study is the first, to our knowledge, to explore empirically the linear and nonlinear impacts of REP on GHG emissions at the microeconomic level in petroleum industry. Findings significant empirical results by measuring empirically the impact of REP on GHG emissions may encourage petroleum companies to consider not only the investment on renewables but also the energy transition path to global net zero emission in their strategy by the med of this century. Second, this study investigate the mediating role of R&D in the relationship between REP and GHG emissions based on the interaction approach introduced by Brambor et al. [9]. Understanding the role of R&D in developing renewable energy and mitigate GHG emissions provides valuable information to police makers in IPCs to advocate this path. Third, we utilize GHG emissions, which consists of Kyoto protocol gases, while CO_2 dominoed in the previous literature. Accordingly, the results of this study will reflect the effort of the said sample to control and reduce other pollution elements, including methane leakage and fugitive. Finally, this empirical study opens a new avenue for academics to further investigate the impact of REP on GHG emissions in petroleum industry under energy transition.

In addition to the introduction in Sect. 1, this paper is organised as follows: Sect. 2 contains information about the theoretical background of the subject. Section 3 is the literature review focuses on the impact of REP on GHG emissions. Section 4 covers the data description and methodology used in the study, Sect. 5 contains empirical results and discussion and Sect. 6 draw the conclusions.

2 Theoretical foundation

Historically, the economic growth based on natural resources were the driving force behind economic growth. However, the turning point on the track of economic growth emerged when the damage to the environment was recognised as one of the most serious problems (Meadows et al. 1972; Brundtland 1987). Subsequently, Porter Hypothesis

(PH), an interesting environmental theory emerged and it postulated that the inclusion of the environment improvement within a company's strategy such as REP and R&D will enhance its environmental and financial performance, resulting in a win–win scenario [53]. Since then, the environment has become the cornerstone of most policies and strategies at the macroeconomic and microeconomic level, particularly in developed economics. The International cooperation in this regard started with the land mark Stockholm Conference of 1972. Subsequently, the United Nations framework convention on climate change was formulated in 1992 to reduce GHG emissions at a level that would prevent dangerous of global warming. The efforts of this body in international climate negotiations resulted in the Kyoto Protocol (1997) and more recently the PA (2015). The latter's ambition goal of net zero emission by 2050 have led many corporations, including IPCs to adopt this goal. In this regard, IPCs in the chosen sample have established the long-term net zero emission, in addition to medium-term targets by 2030, focusing on both REP and GHG emissions reduction. For instance, based on corporate sustainability reports, announced 2030 renewable installation targets include Eni (15 GW), Repsol (18 GW), Equinor (12 GW), and ONGC (10 GW), INPEX (3), Hellicin petroleum (2) and Saras (1). Concurrently, significant scope 1 and 2 GHG emissions reduction targets have been set, for instance, Equinor, Shell, Hellenic Petroleum, Eni, and BP are targeting a 50% reduction by 2030, in the baseline of 2015, 2016, 2017, 2018 and 2019, respectively. Other commitments include a 40% reduction by Total Energies and Galp on the base of 2015 and 2017, respectably, a 30% reduction by OMV, and a 25% reduction by Petronas and Ecopetrol in the baseline year of 2019. Similarly, Lukoil and Repsol aim for a 20% reduction, while Bangkok Oil aims for carbon neutrality by 2030.

Beyond broad GHG emissions goal, some IPCs companies from the said sample have established more granular targets focusing on specific emissions Like methane metrics. Specifically, commitments include a 30% reduction by YPF (baseline 2019) and a 4% reduction by Novatek (baseline 2021) by 2030. More ambitiously, Occidental Petroleum aims to eliminate routine methane flaring in its upstream operations by 2030. A similar trend is observed in the adoption of zero routine flaring by 2030, for example, YPF, Petronas, Galp, LuKOIL, Shell, and Total Energies, in the said sample, are committed to eliminate routine flaring in their operations by 2030.

In the net zero emission path also, twelve leading petroleum companies including PB, Eni, Enqier, OXY, Shell, Repsol and total energy formed the Oil and Gas Climate Initiative (OGCI) in 2014. This initiative aims to accelerate the transition to a low-carbon future by investing in R&D. The OGCI's 2023 Progress Report documented a collective investment of $65.8 billion in low-carbon technologies between 2017 and 2022. This investment has driven significant operational improvements in terms of emissions, including a 45% reduction in routine flaring, a 50% cut in upstream methane emissions, and a 21% decrease in upstream carbon intensity over the same period. Furthermore, the initiative fosters knowledge sharing with non-member petroleum companies to amplify global decarbonization efforts achieve net zero emissions. All ambitious metrics and initiatives aforementioned reflect the efforts of IPCs in achieving net zero emissions by 2050. However, many uncertainties remain about the progress of the energy transition and whether it will be sufficiently rapid to prevent the growing severe impacts of global warning.

3 Literature review

Since the beginning of 1970, environmental degradation has been widely investigated (Saleh et al. 2024). However, this literature review aims to scrutinise the impact of REP on GHG emissions. The commonality among these studies is the use of a qualitative methodology and the provision of various conclusions. Qualitative studies can be divided into two categories, groped and individual studies.

3.1 Groped studies

Groped studies mean studies' samples consist of many companies. For example, Hansen and Steen [20] conducted 23 interviews with representatives from 20 Norwegian offshore petroleum companies to determine corporate motivations for engaging in offshore wind electricity generation. The results revealed that a huge potential market opportunity was the major motivation for developing and exploiting current resources. In a follow-up study, Nasiritousi [49] scrutinised the climate change activities of the largest 10 IPCs based on the daily production of hydrocarbons. The results revealed that all petroleum companies have already adopted mitigation measures of GHG emissions in their operations. However, investor-owned companies based in developed countries encounter more pressure to take action from shareholders and NGOs than companies in developing countries. Similar results were found by Zhong and Bazilian [71] who analysed the progress of Statoil, Shell, Total SA, BP, Chevron, ExxonMobil and Eni in promoting solar, wind and biofuels. Their results showed a diversity of strategies for scaling up renewables, leveraging their expertise in oil and gas production and creating an integrated value chain. In addition, clean energy solutions including battery storage, power plants, hardware integrators, equipment leasing, software were supported by the sampled companies. Nevertheless, the new models that petroleum companies adopted in the sample are still in its fancy. Graham [18] also investigated the renewable energy deployment in 10 Canadian petroleum companies, namely Suncor, Tran Canad, Teck, Husky, Enbridge, Fortis, Imperial Oil, Cenovus, Canadian Natural Resources Limited and Parkland Fuel. Results revealed a significant investment in the first six petroleum companies. The other petroleum companies did not invest in renewable energy during the study period. In the same year, Boon [8] analysed the decarbonisation strategy of the energy sector in a sample of 10 EU and U.S. oil and gas companies, namely ExxonMobil, Royal Dutch, Shell, Chevron, Total, BP, Equinor, ConocoPhillips and Eni. EU oil and gas companies are in a better position in terms of energy transition than U.S. oil and gas companies because EU petroleum companies have expanded their investments in renewable energy, started reporting on their sustainability efforts and voluntarily committed to reducing their GHG emissions. In the same path, [42] analysed the green energy transition path of nine major petroleum companies concerning their transition targets, actions and investments. The results revealed that petroleum companies began the green energy transition by increasing investments in renewable energy as a long-term strategy, and some chose to concentrate on energy efficiency. The study also reported that petroleum companies were focusing on geothermal, solar and wind energy and have acquired experience in the application of geothermal energy. In the beginning of this decades [59], analysed the correlation between the operational indicators of 18 major oil and gas companies and national energy transition strategies. On the whole, the results revealed a weak correlation between the two variables. Nevertheless, major oil and gas companies in the EU

have increased their investments in renewable energy and are more likely than others to adapt their businesses to the energy transition as a result of strict environmental regulations. From another avenue, [41] analysed the role of R&D in the energy transition of 18 major petroleum companies. The results reveal that high R&D investments and innovation were highly associated with energy transitions. [44] investigated the strategies and perceptions of 11 major petroleum companies about the energy transition, the results revealed that major petroleum companies were attentive to climate change and had the capacity to transition to the use of green energy. Finally, [11] examined the economic and social–political pressure on 12 EU petroleum companies in addition to their response to take action against climate change between 2005 and 2019. The results revealed that the pressures were insufficient to reinforce these petroleum companies to abate their fossil fuel investments in favour of green energy to accelerate the green energy transition.

3.2 Individual studies

The second category of quantitative studies is individual studies. These are studies in which the sample is only one company. For example, [51] explored the environmental strategic decisions of Statoil concerning investments in conventional and green energy, results revealed that new initiatives were taken to close the gap between Statoil's conventional and green energy paths within offshore wind and CCUS. The company is highly dependent on hydrocarbon and will rely on it in the coming years because it accounts for 80% of the company's new investments. In a subsequent study [1] used DONG-Ørsted as benchmark, which changed its business model totally to become a renewable energy company, for comparison to petroleum majors. Results revealed that major petroleum companies were not doing well in terms of their energy transition to renewable energy. In a recent study, Toghyani and Saadat [61] explored the validity of incorporating wind, solar, and biomass energy alongside the existing natural gas power plant and grid connection on Esfahan oil refinery company plant to meet the increasing energy demands of the refinery. The results indicate that the integration of renewable energy sources can lead to a 44.5% reduction in CO_2.

As mentioned above, there is a scarcity of quantitative studies in the related literature. To be specific, there are only two empirical studies related to our subject. First, Jarboui [28] investigated the effect of RE on operational and environmental efficiency in a sample of 45 U.S. petroleum companies from 2000 to 2018, using a true fixed effect model, the results reveal that renewable energy enhanced operational and environmental efficiency of petroleum companies by reducing CO_2 emissions. In general, the findings highlight that U.S. petroleum companies have begun their transition to reduce their CO_2 emissions. In the second study, Fethi and Rahuma [17] examined the dynamic impact of eco-innovation (i.e., investment in physical assets, training and R&D on CO_2 emission reductions in a sample of 17 petroleum companies between 2005 and 2016, utilising ARDL, the results reveal that investment in physical assets significantly reduced CO_2 emissions in the long term, whereas R&D and training made a significant reductions to CO_2 emissions in the short term. This study differs from the two aforementioned studies in many ways. First, this study measures the linear and the nonlinear relationship of the impact of REP on GHG emissions. Second, this study utilizes the dynamic GMM model and covers a unique period starting from the ratification of PA (i.e., 2015). Third, the sample of this study consists of only IPCs in the upstream sector. Fourth, we include

REP in this study while REP was not included in the model used in the study of Fethi and Rahuma [17].

Before we close this section, it is worth mentioning that there is a plethora of empirical studies that seek to explore the impact of REP on the environment at macroeconomics level. For example, using different samples of countries [12, 30, 31, 46, 55, 62] found a significant effect from the side of renewable energy on the environmental quality.

In summary, even though the investigated literature enriched our understanding of renewable energy deployment in petroleum companies, the authors of these studies relied on qualitative assessments of strategic documents of petroleum companies rather than looking at how much actual REP affects GHG emissions, so the findings remain inconclusive. Thus, it is imperative to mention that, to the best of our knowledge, this is the first empirical study to measure the impact of REP on GHG emissions in the upstream sector. Besides, existing studies fail to provide decisive, tangible recommendations for reducing GHG emissions through REP in petroleum companies. In response to the calls of [2, 38], we attempted to fill the gap in the relevant Literature by measuring the impact of actual REP on GHG emissions in IPCs. Therefore, this paper is a novel contribution to the related Literature because it includes measurements of the impacts of REP on GHG emissions, thereby providing thoughts and insights to petroleum company managers in their plans of perusing net-zero emissions by 2050.

4 Data and research methodology

4.1 Data description

Annual data covering 2015–2023 for 27 IPCs are utilized in this study. The starting date of the study was selected because of the PA ratification in 2015. Clearly, the majority of petroleum companies in the sample had set net-zero emissions goals and started deploying renewables since the ratification of PA [24]. The ending date was chosen because of data availability. IPCs was a good choice to study because they are a major contributor to global GHG emissions. For example, Total, Lukoil and Eneos produced about 44, 48.5 and 25.5 million tonnes of CO_2 equivalent respectively in 2019 (Annual reports, 2019). In addition, the huge profits IPCs gain enable them to invest in renewables. Petroleum companies are also obliged to disclose their environmental, social and governance data according to the non-financial reporting directive (2014/95/EU) and annual sustainability reports (EU 2023/2772).

The data used in our study was extracted from the annual reports of the studied companies. These companies include Shell, Eni S.p.A, Total, Repsol, Gazprom, Equinor, Hellenic Energy, Eneos, Inpex, OMV Aktiengesellschaft, BP, Ecopetrol, Rosneft, Galp, Lukoil, Novatek, Suncor Energy, Vermilion Energy, Saras, Occidental, Petroleum National Berhad, YPF Sociedad, Bangchak, Gail NS, Indiaoil Corporation limited, Oil India Limited and ONGC. These IPCs were chosen based on the following inclusion criteria. Each company must: (i) have a market capitalisation of more than US$1 billion; (ii) be heavily engaged in the upstream sector; (iii) have a self-generation of REP such as solar, wind, hydro, biomass and geothermal with related annually reported data; (iv) have a vision of net zero emission; (v) be listed on a public stock exchange.

The variables used in this study included GHG emissions is in line with the work of Hailemariam et al. [19]. REP is in line with [28, 32]. R&D is in line with the work of [70]. Total assets (TA) is in line with [28]. These variables are classified as follows: the

Table 1 Variables, measurements and sources

Variables	Measurement	Source
GHG emissions	Thousands of tonnes of CO_2 equivalent	Annual sustainability reports
REP	GWh	Annual sustainability reports
R&D	Million dollars	Annual financial reports
TA	Million dollars	Annual financial reports
Employees	Numeric number	Annual reports

Table 2 Descriptive statistics

Variables	Obs	Mean	Std. Dev	Min	Max
GHG emissions	243	−0.05	0.95	−3.17	2.52
REP	243	0.05	0.94	−2.64	2.66
R&D	243	−0.01	0.90	−2.22	2.53
TA	243	0.00	0.94	−2.18	2.16
Employees	243	0.00	0.94	−2.64	2.62

Table 3 Corelation matrix

Variables	GHG	REP	R&D	TA	Employees
GHG	1.000				
REP	−0.211	1.000			
R&D	0.131	0.087	1.000		
TA	−0.056	0.402	0.256	1.000	
Employees	0.182	0.078	0.223	0.132	1.000

dependent variable is GHG emissions, the endogenous variable is the one–year-lagged GHG emissions, the explanatory variables are REP and R&D while TA is a control variable and serves as the IV for the system-GMM estimation. All data were standardised so that the same scale was used for all variables. Table 1 presents the definition, measurement unit and sources of all variables. Descriptive statistics are shown in Table 2 while Table 3 contains the correlation matrix.

Table 2 shows that the mean value of GHG emissions is – 0.037 with a maximum value of 2.199. GHG emissions are calculated by aggregating the values of Scope 1 and Scope 2 GHG emissions. Scope 2 is added to scope 1 in our analysis because many of IPCs consume part of their REP internally, for example Ecopetrol and Bangchak's consumption from self-generated REP was around 54 GWh and 23 MWh respectively in 2021. The mean value of REP is 0.11 with maximum and minimum values of 2.423 and -2.455, respectively. REP is calculated based on the generation of electricity in GWh annually. The data is standardised to minimise heterogeneity that arises from the different sizes of IPCs.

Table 3 contains the correlation matrix. The unconditional correlations between the variables are generally low. The highest correlations were observed between REP and TA (0.40) and between R&D and TA (0.25), suggesting that large petroleum companies produce more RE and spend more on R&D. Besides, the amount of REP was found to be law and negatively correlated (-0.21) to GHG emissions. Conversely, the correlation between R&D and GHG emissions was positive and very Law. Firms with many employees had higher GHG emissions. For the other correlations, the absolute value of correlation was below 0.40. Thus, multicollinearity was not a problem in our estimations.

4.2 Methodology

In this study, generalised method of momentum (GMM) approach was applied to measure the impact of REP on GHG emissions. It was proposed by Holtz-Eakin et al. [23] and subsequently developed by [6, 7, 58]. GMM is a panel dynamic data method estimates parameters in models, and it is used when the number of cross-sectional units is larger than the period of time in a panel [58]. Hence, the number of cross-sections (N) is 27 while the period (T) is 9. Besides, this method can be used to control the endogeneity of the lagged independent variable, the bias of omitted variables [50], measurement errors and unobserved panel heterogeneity.

The GMM method involves utilising the first-differenced estimator to evaluate regression equations in first differences using lagged levels of independent variables as instruments. This method eliminates company-specific fixed effects, thereby addressing the potential exclusion of time-invariant company-specific factors that might impact GHG emissions. However, the first-differenced GMM estimator, as introduced by Arellano and Bond in 1991, is not suitable for use when time series are highly persistent and the number of time series observations is limited. In such situations, lagged levels of independent variables are weak instruments for subsequent first differences, resulting in biased outcomes. Arellano and Bover [7] and Blundell and Bond (1998) recommended using the system-GMM technique to resolve this issue, which combines regression equations for first differences and levels into a single system. The level equations use lagged first differences of the series as instruments. Therefore, we begin with the general model.

$$yit = aYit - 1 + BXit + ni + \varepsilon it \text{ for } i = 1, 2 \ldots, \mathrm{N} \text{ and } t = 1, 2, \ldots, \mathrm{T}, \tag{1}$$

where i represents the company index, t denotes the time index, y is the dependent variable (GHG), X is a vector of other conditional variables affecting GHG, iη captures unobserved company-specific effect terms while εit indicates the idiosyncratic disturbance term. Arellano and Bond [6] recommend transforming the model into first differences to eliminate company-specific effects.

$$yit - yi, t_{-1} = a\left(yit_{-1} - yi, t_{_2}\right) + B\left(xit - xit_{-1}\right) + \left(\mathcal{E}it - \mathcal{E}i, t_{-1}\right) \tag{2}$$

GMM estimators are typically used in one-step and two-step variants [6]. The one-step estimator uses weighting matrices that are independent of the estimated parameters. In contrast, the two-step GMM estimator uses optimal weighting matrices for which the moment conditions are weighted by a consistent estimate of their covariance matrix. Consequently, the two-step estimator is generally more efficient than the one-step estimator. However, using the two-step estimator in small samples has notable drawbacks. These drawbacks stem from the proliferation of instruments. Through simulation analysis [64], demonstrated that employing many instruments in two-step GMM estimation can lead to biased standard errors and parameter estimates.

Roodman [58] suggested reducing the dimensionality of the IV matrix to address issues caused by the proliferation of instruments. We adopted this approach in this study, employing the moment conditions and using the two-step estimator. We lowered the dimensionality of the IV matrix based on Roodman's recommendations. The prime objective of this study is to measure the impact of REP on GHG emissions. To do so, we followed PH (i.e., REP improves a company's operations and decreases GHG emissions). Many studies had the same perspective, such as that of [28, 33], the results of which

suggest that increasing REP improves the quality of the natural environment by reducing GHG emissions. Based on this discussion, we expect REP to positively impact GHG emissions reduction attempts.

$$GHGit = ai + B1GHG_{t-1} + \beta 2\,(\mathrm{REPit}) + \beta 3\,(\mathrm{R\&Dit}) + \beta 4\,(\mathrm{TAit}) + \mu i + \delta t + \varepsilon i, \mathrm{t} \tag{3}$$

where GHGit is an indicator for environmental emissions in company I at year t, i is the number of cross-sections (= 1,…,N), t is the number of time series (= 1,…,T), REPit denoted renewable energy production, *R&D*it denoted research and development expenses, TA represents total assets, μi is the unobserved company-specific fixed effects, δ*t* is the time trend and ε*i*, t is the error term.

Equation 1 is for only the short-run effects. Additionally, to align this study with the previous literature on this topic, R&D was added to the model as a second explanatory variable to assess its impact on GHG emissions in line with the method of [17] who measured the impact of R&D on CO_2 emission within 17 petroleum companies. R&D is related to innovation and technology for patents which are useful to improve the existing technology in order to reduce GHG emissions [12]. We formulated the following equation to explore the nonlinear relationship between both REP, R&D and GHG emissions:

$$GHGit = \alpha i + B1GHG_{t-1} + \beta 2\,(\mathrm{REPit}) + \beta 3\,(\mathrm{REPit})^2 + \beta 4\,(\mathrm{R\&Dit}) + \beta 5\,(\mathrm{R\&Dit})^2 + \beta 6\,(\mathrm{TAit}) + \mu i + \delta t + \varepsilon i, \mathrm{t} \tag{4}$$

where $(\mathrm{REPit})^2$ and $(\mathrm{R\&Dit})^2$ are the square terms of REP and R&D respectively. The squared terms of REP and R&D were taken to measure the effect of both variables on GHG emissions at the initial stage and in a later stage. Thus, if the inverted U-shaped holds for REP and R&D, then the signs of B2 and B4 related to REP and R&D are positive, whereas B3 and B5 related to REP squared and R&D squared are negative.

The first equation was formulated to calculate the short-run parameters. To measure the long-run relationship between variables in Eq. 1, we generated the long-run coefficients by adopting the following Eqs. 5 and 6:

$$\mathrm{REP} \div [1 - \emptyset] \tag{5}$$

$$\mathrm{R\&D} \div [1 - \emptyset] \tag{6}$$

where REP and R&D are the coefficients of the short-run parameters and ϕ is the coefficient of the endogenous variable in the model of Eq. 1.

The role of R&D investment in developing REP to enhance the latter's impact on GHG emissions is vital [19]. Therefore, we expect R&D investment to play a mediating role in the relationship between REP and GHG emissions in line with [36]. To do so, Eq. 7 is extended to capture the interaction between REP and R&D in one hand, and REP squared and R&D on the other hand.

$$D\mathrm{GHGit} = \alpha i + B1GHG_{t-1} + \beta 2\,(\mathrm{REPit}) + \beta 3\,(\mathrm{REPit})^2 + \beta 4\,(\mathrm{R\&Dit}) + \beta 5\,(\mathrm{REPit*R\&D}) + \beta 6(\mathrm{REPit*R\&D}) + (\mathrm{TAit}) + \mu i + \varepsilon i, \mathrm{t} \tag{7}$$

where: (REPit*R&D) is REP multiplied by R&D; ($\mathrm{REP^2it}$*R&D) is the exponential of REP multiplied by R&D.

5 Empirical results and discussion

Table 4 presents the two-system GMM estimator and the two-step corrected System-GMM estimator results of Eg 3. First, it is notable that the lagged dependent variable is statistically significant at a 1% level, confirming that a dynamic model is justified in this analysis. The positive and significant singe of the lagged variable (GHG_{t-1}) indicates that a one standard deviation in lagged GHG emission results in a 0.51% rise in GHG emissions. This implies that a rise in previous GHG emissions would heighten current GHG emissions in the studied IPCs. This finding aligns with those in global reports that showed there were rapid increases in GHG emissions in the past few years due to petroleum extraction [13]. Secondly, REP, the core variable of this model, was found to have a negative and statistically significant Link with GHG emissions at a 1% level, as well as for the system-GMM robust. As such, a one standard deviation increase in REP will result in a 0.36% drop in GHG emissions in the short run at the 1% level of significance *ceteris paribus*. The negative coefficient implies that REP is a vital factor in reducing GHG emissions because the production process for RE emits very little GHG emissions. Accordingly, IPCs pursuing REP are likely to improve their environmental performance. This outcome aligns with that of [28] who used the true fixed effect model in a sample of 45 USA oil and gas companies. Similar results were also achieved by [15, 40] who used the GMM methodology but at the level of macroeconomics. This result is also in line with [47, 48, 56, 57, 60] who used second generation econometric methods such NRDL Distributed Lag-Pooled Mean Group, Autoregressive Distributed Lag-Dynamic Fixed Effects, Fully Modified Ordinary Least Squares and Autoregressive Distributed Lag-Error Correction Model and QARDL. Importantly, these results serve SDG7 and 13. (i.e., affordable and Clean Energy and climate action. Thirdly, R&D was found to have a statistically significant and negative relationship with GHG emissions. Thus, a one standard deviation in R&D causes a 0.13% drop in GHG emissions in the short run at the 10% level of significance *ceteris paribus*. This outcome is congruent with those of [17, 37] while it contradicts those of [12] who concluded that the development of a patent did not lead to a significant reduction in CO_2 in 35 OECD countries. Fourth, the IV included in the model posits a positive and statistically significant connection with GHG emissions. As such, a one standard deviation increase of TA leads to a 0.16% increase in GHG emissions.

Table 4 The impact of REP and R&D on GHG emissions

Variables	Two-step system GMM	Two-step system-GMM (robust)
GHGit-1	**0.628*** ** (10.34)	**0.628*** ** (3.49)
REP	**−0.361*** ** (−7.86)	**−0.361*** ** (−3.04)
R&D	**−0.134*** (−1.98)	**−0.134** (−0.79)
TA	0.165*** (3.70)	0.165 (1.44)
Constant	−0.016 (−0.62)	−0.016 (−0.24)
Sargan test	0.256	0.256
Hansen test	0.122	0.122
AR (1)	0.003	0.013
AR (2)	0.542	0.610
Number of groups	27	27
Number of instruments	23	23
Observations	189	189

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

To conclude, the results suggest that REP and R&D is essential in reducing GHG emissions at the company level. These outcomes resonate with our earlier claims that promoting REP and R&D lower GHG emission. The contributions of REP and R&D to GHG emissions reduction efforts significantly differ, with contributions ranging from − 0.36% for REP to − 0.13% for R&D. The much larger contribution of REP than R&D may be attributed to the fact that the IPCs started greatly investing in RE projects after setting the goal of net-zero emissions after 2015 [42]. While IPCs was concentrating on R&D before 2015 to control GHG emissions. Hence, our results support the PH that inspiring environmental investment has positive effects on EP by decreasing GHG emissions in the studied sample [54].

Table 5 contains the result of the nonlinear relationship of Eq. 4. The result reveals that REP has a negative impact on GHG emission reduction while its quadratic has a positive impact on GHG emission reductions. Of note, the coefficients are the same with regard to the system-GMM robust but at the level of 5%. This means that REP does not affect GHG emissions reduction efforts at the beginning, however, the effect becomes influential in reducing GHG emissions above a certain threshold, indicating that a U-shaped pattern is valid in this relationship. This finding aligns with that of [12] who found an inverted U-shape from the side of RE on CO_2 in a sample of 35 OECD countries. Also, this result is in line with [16] who found a threshold effect between renewables and carbon emission efficiency within 32 developed countries. Similarly, R&D has the same trend as REP on GHG emission reductions. Specifically, the impact of R&D on GHG emissions reductions is negative in the initial stages. However, as long as IPCs spend more on R&D, this effect becomes positive in reducing GHG emissions above a certain threshold, indicating that a U-shaped relationship exists between the two variables. This

Table 5 The nonlinear relationship between REP and R&D on GHG emissions

Variables	Two-step system-GMM	Two-step system-GMM (robust)
GHGit-1	**0.619*** **	**0.619*** **
	(3.55)	(3.50)
REP	**0.589*** **	**0.589** **
	(3.55)	(2.61)
REP^2	**− 0.635** **	**− 0.635** **
	(− 3.17)	(− 2.48)
R&D	**0.214**	**0.214**
	(1.26)	0.93
$R\&D^2$	**− 0.334** **	**− 0.334** **
	(− 3.00)	(− 2.31)
TA	0.143	0.143
	1.10	0.93
Constant	0.434**	0.434**
	)3.59(	2.29
Sargan test	0.744	0.744
Hansen test	0.711	0.711
AR (1)	0.004	0.005
AR (2)	0.831	0.836
Number of groups	27	27
Number of instruments	18	18
Observations	162	162

Parametric Estimates for the U-shaped inverted relationship between REP and R&D on GHG emissions, controlling for TA. (Note: t-statistics in parentheses; ***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively. Estimations made using xtabond2 routine in Stata)

Table 6 Long-run effect of REP on GHG emissions

Variables	Two-step system GMM (robust)
REP	**−0.974 **** (−2.23)
R&D	**−0.361** (−0.64)

Parametric Estimates for coefficients of REP and R&D in the long run. (Note: t-statistics in parentheses; ***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively. Estimations made using xtabond2 routine in Stata.)

Table 7 Role of R&D in mediating the relationship between REP and GHG emissions

Variables	Two-step system-GMM	Two-step system-GMM (robust)
GHGit-1	**0.768*****	**0.768*****
	(7.51)	(3.36)
REP	**0.238****	**0.238**
	(2.45)	(1.16)
REP^2	**−0.708*****	**−0.708****
	(−5.36)	(−2.68)
REP × R&D	**0.7062*****	**0.706***
	(3.08)	(1.74)
REP^2 × R&D	**−0.478*****	**−0.478***
	(−2.88)	(−1.71)
R&D	0.447*	0.447
	(1.70)	(0.84)
TA	0.017	0.017
	(0.19)	(0.08)
Constant	0.488***	0.488
Sargan test	(4.18)	(1.84)
	0.850	0.850
Hansen test	0.319	0.319
AR (1)	0.009	0.021
AR (2)	0.384	0.446
Number of groups	27	27
Number of instruments	20	20
Observations	216	216

Parametric Estimates for the role of R&D in mediating the relationship between REP and GHG emissions, Controlling for TA. (Note: t-statistics in parentheses; ***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively. Estimations made using xtabond2 routine in Stata.)

outcome resonates with several existing studies concerning the inverted U-shape trend between R&D and GHG emissions [14, 15, 69].

Table 6 shows the long relationship results from the side of REP and R&D on GHG emissions of Eq. 5 and Eq. 6 as it was in the short run. The findings reveal a positive relationship between REP and GHG emission reduction in IPCs in the long run. However, the findings also revealed that R&D does not have a long impact on R&D. This findings aligns with that of [17] who claimed that R&D did not have a long relationship with CO_2 emissions in a sample of 17 IPCs. This finding is also consistent with those of [55] at the macroeconomic level. IPCs might have been investing hugely on renewable project since the ratification of the PA in 2015.

According to the U-shaped relationship between REP and GHG emissions, we proceeded to use Eq. 7 to capture the role of R&D in mediating the relationship between REP on GHG emissions (Table 7).

The model of Eq. 7 incorporates the interaction term between REP and R&D (REP × R&D) and REP squared and R&D (REP^2 × R&D). Based on [34], it is not of intent to interpret the individual term of the independent variable and its squared and

mediator values, so these coefficients are not of interest. Interestingly, results show that the coefficients associated with REP and R&D interaction are positive, and the interaction between REP squared and R&D is negative. Besides, both interaction terms are statistically significant determinants of REP. These empirical results indicate that the inverted U-shaped relationship between both REP and R&D play a vital role in mediating the effect of R&D in the REP nexus. Clearly, when the interaction terms (i.e., $(REP \times R\&D)$ and $(REP^2 \times R\&D)$ have positive and negative signs, respectively, and are significant, the effect of REP on GHG emissions increases with R&D, and further REP has a vanishing effect on GHG emission reductions. This result implies that the effect of REP on GHG emission reduction increases monotonically with R&D. Put differently, the positive effect of REP on GHG emissions reduction declines as R&D decreases. This finding aligns with that of [19], who found that R&D investment in renewable technologies has a positive effect on environmental quality by reducing GHG emissions in advanced and emerging economies. Therefore, our findings support the view that R&D embedded within REP is effective tool in reducing GHG emissions.

6 Robustness test

In this section, the authors investigate whether the results of the previous analysis are robust. We further test the GMM model by replacing TA as IV in Eqs. 3, 4, 5, 6, and 7 with employees, in line with the work of [28, 66]. The empirical results about the replaced IV show similar results. Specifically, REP and R&D have a negative effect on GHG emissions, as shown in Table 8. While in Table 9, the sing of the REP and R&D is positive, whereas their quadradic values are negative, which indicates that there is an inverted U-shaped on the impact of REP and R&D on GGH emissions. In Table 10, the sings for both the REP and R&D in the long-run relationship are negative but it is significant only for REP at the 1% level. Finally, Table 11 show that the two interaction terms have positive and negative signs and are significant. To conclude, the empirical results with the replaced IV estimator (employees) of all equations (from 4 to 7) have the same signs as those obtained with TA IV.

Table 8 The impact of REP and R&D on GHG emissions

Variables	Two-step system-GMM	Two-step system-GMM (Robust)
GHG-1	**0.575** ***	**0.575** ***
	(6.98)	(3.06)
REP	**−0.381** ***	**−0.381** **
	(−9.18)	(−2.96)
R&D	**−0.191** ***	−0.191
	(−4.02)	(−1.28)
Employees	0.259 ***	0.259**
	(5.28)	(2.43)
Constant	0.028	0.028
	(1.24)	0.46
Sargan test	0.221	0.221
Hansen test	0.149	0.149
AR (1)	0.005	0.018
AR (2)	0.51	0.579
Number of groups	27	27
Number of instruments	23	23
Observations	189	189

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

Table 9 Nonlinear relationship between R&D, REP and GHG emissions

Variables	Two-step system-GMM	Two-step system-GMM (Robust)
GHG-1	**0.554 *****	**0.554 *****
	(3.96)	(3.55)
REP	**0.538 *****	**0.538 ****
	(2.72)	(2.05)
REP^2	**−0.624 *****	**−0.624 ****
	(−2.97)	(−2.55)
R&D	**0.193**	**0.193**
	−1.41	0.80
$R\&D^2$	**−0.324 ****	**−0.324 ****
	(−2.68)	(−2.44)
Employees	0.145 **	0.145
	2.07	1.37
Constant	0.489***	0.489 **
	)3.91(	2.82
Sargan test	0.784	0.784
Hansen test	0.802	0.802
AR (1)	0.006	0.007
AR (2)	0.739	0.763
Number of groups	27	27
Number of instruments	18	18
Observations	162	162

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

Table 10 Long-run effect of REP and R&D on GHG emissions

Variables	Two-step system GMM (robust)
REP	**−0.898*****(−2.72)
R&D	**−0.451** (−0.91)

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

6.1 Heterogeneous effects

To assess the homogeneity of our results, we conduct further empirical analysis at the regional level concentrating on two subsamples, namely: EU IPCs and non-EU IPCs [35]. The results presented in Table 12, reveal significant regional disparities. In the short run, the impact of REP on GHG emission is most significant for EU IPCs than non-EU IPCs. Similarly, REP's coefficient value is substantially larger for EU IPCs compared to their non-EU counterparts. Interestingly, this pattern persists in the long run, where the impact of REP on GHG emissions reduction remains both more significant and of a greater magnitude for EU IPCs compared to their non-EU counterparts, see Table 13.

We attribute the superior effectiveness of EU IPCs concerning the impact of REP on GHG emissions to several facts. First and for most is the stringent regulatory frameworks within the European Union Like the EU green deal which mandates climate neutrality by 2050 and a minimum 55% reduction in GHG emissions by 2030 based on 1990 levels. Second, EU IPCs have demonstrated a purely concentrated focus on renewable energy, while many of the non-EU IPCs continue to utilize natural gas as a primary component of their decarbonization strategy such as Gail India, Petroliam Nasional Berhad and ONGC. Last, the capacity of REP is generally larger in the EU IPCs. To prove, we further divided the aggregate sample into two groups on the ground of REP scale. We not only observe that IPCs with higher REP capacity exhibit more pronounced GHG emissions reductions but also the majority of the EU IPCs predominantly constitute the

Table 11 Role of R&D in mediating the relationship between REP and GHG emissions

Variables	Two-step system-GMM	Two-step system-GMM (Robust)
GHG-1	**0.691*** **	**0.691** **
	(7.13)	(2.65)
REP	**0.216** **	**0.216**
	(2.27)	(1.36)
REP^2	**−0.729*** **	**−0.729** **
	(−5.71)	(−3.80)
REP*R&D	**0.574** **	**0.574**
	(2.67)	(1.57)
REP^2*R&D	**−0.402** **	**−0.402**
	(−2.61)	(−1.54)
R&D	0.514**	0.514
	(2.12)	(1.51)
Employees	0.087	0.087
	(1.66)	(1.16)
Constant	0.517***	0.517**
	(4.80)	(2.62)
Sargan test	0.824	0.824
Hansen test	0.507	0.507
AR (1)	0.452	0.057
AR (2)	0.015	0.531
Number of groups	27	27
Number of instruments	20	20
Observations	216	216

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

Table 12 The impact of REP and R&D on GHG emissions (Subsample)

Variables	EU IPCs two-step system GMM	EU IPCs two-step system GMM (robust)	Non-EU IPCs two-step system GMM	Non-EU IPCs two-step system-GMM (robust)
GHGit-1	**0.466*** ** (3.36)	**0.466** (1.79)	**0.434*** ** (3.68)	**0.434** (1.59)
REP	**−0.407*** ** (−8.04)	**−0.407*** ** (−3.86)	**−0.168** ** (−2.20)	**−0.168** (−0.87)
R&D	**−0.124** (−0.94)	**−0.124** (−0.34)	**−0.057** (−0.59)	**−0.057** (−0.22)
TA	**0.166** (2.05)	**0.166** (0.96)	**0.175** (1.30)	**0.175** 0.64
Constant	−0.103 (−1.93)	−0.103 (−0.64)	0.015 (0.22)	0.015 (0.10)
Sargan test	0.621	0.621	0.173	0.173
Hansen test	0.105	0.105	0.170	0.170
AR (1)	0.070	0.106	0.063	0.156
AR (2)	0.842	0.860	0.550	0.638
Number of groups	14	14	13	13
Number of instruments	12	12	12	12
Observations	98	98	91	91

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

Table 13 Long-run effect of REP and R&D on GHG emissions (subsample)

Variables	EU IPCs two-step system-GMM (Robust)	Non-EU IPCs two-step system-GMM (robust)
REP	**−0.763*** ** (−4.51)	**−0.297** ** (−2.25)
R&D	**−0.233** (−0.80)	**−0.101** (−0.61)

***p < 0.01, **p < 0.05, *p < 0.1 indicate significance at 1%, 5% and 10% respectively

high-capacity group. With respect to the results of R&D, its coefficient is significant only in the short run for both groups. Importantly, the impact of R&D on GHG emission is significantly weaker than the impact of REP on GHG emission both in short and long run. To conclude, our findings underscore the significance of regional characteristics in the relationship between REP and GHG emissions, where the effect of REP on GHG emissions is found to be more preannounced in EU IPCs relatively to non-EU IPCs. All the results from the subsample analysis are in consistent with the earlier results related to the aggregate sample.

Before closing the empirical side, it is noteworthy to mention that the reliability of the GMM depends on satisfying a few conditions [6]. Specifically, the two-step system GMM robust indicates that there is no heteroscedasticity problem in the results. Also, there is no autocorrelation for the second-order Auto-regression in the first differences. Besides, the number of groups exceed the number of instruments so that the rules of thumb condition has met in our analysis. Finally, the Sargan and Hansen statistic are more than 0.1. As such, we can conclude that the results of this study are valid and reliable since these conditions have been met.

7 Conclusion and policy implications

The increasing level of GHG emissions due to oil and gas extraction and production by petroleum companies pose one of the most serious threats to the environment. The rapid growth of renewable energy provides petroleum companies a cross cutting solution to mitigate GHG emissions and meet the net zero emission goal by the mid of this century. However, how REP impacts GHG emissions in petroleum industry under energy transition, remain to be investigated. Therefore, the prime aim of this paper is to empirically explore the impact of REP on GHG emissions under energy transition in a panel consist of 27 IPCs over the period from 2015 to 2023. The dynamic system GMM method was utilized in the framework of PH. We find strong evidence that REP positively affects GHG emissions reduction in the short and long run. The outcomes also validate an inverted U-shape relationship between REP and GHG in the sense that REP does not affect GHG emission at first stage, but after a certain threshold, REP positively affects GHG emission. Furthermore, findings revealed that R&D positively impact GHG emissions reduction only in the short run and this impact has an inverted U-shape trend. Also, R&D is found to plays a vital role in mediating the relationship between REP and GHG emissions reduction. Finally, to address potential heterogeneity, we segmented IPCs into two groups of EU and non-EU, the result underscores the significance of regional characteristics in the relationship between REP and GHG emissions, where the effect of REP on GHG emissions is found to be more preannounced in EU IPCs relatively to non-EU IPCs.

7.1 Policy implications

This study provides valuable information for policymakers in petroleum companies. First, the findings demonstrated that the ability of petroleum companies in reducing GHG emission depends remarkably on the scale of REP. In other words, the impact of REP in reducing GHG emissions is not immediate; in the initial stage, the impact of REP on GHG emissions reduction may be negligible. But above a critical threshold, REP will affect positively GHG emission reduction. Therefore, it is imperative for petroleum

companies to set a long-term plan designed to scale REP beyond this critical threshold in order to positively affect GHG emissions reduction.

Second, R&D is also a vital variable not only in decreasing GHG emissions directly but also in enhancing the impact of REP on GHG emissions reduction. Therefore, we recommend the policymakers at IPCs to dedicate and increase the budget of R&D in two area. Namely: (i) Next-generation renewable like green hydrogen and (ii) low carbon technologies such as CCUS and the direct air capture to reduces GHG emissions from the operational process of petroleum industry. This dual R&D strategy will result in simultaneous increase of REP and the reduction of GHG emissions.

7.2 Limitations of study

The primary Limitation of this study was data availability. More specifically, many IPCs have a self-generation production of renewables but some of these companies do not disclose their annual REP. Hence, the sample of the study is restricted to only 27 IPCs. Thus, we should be cautious to generalize our results to other IPCs. In the future, researchers might widen the sample of the study as more petroleum companies have been increasingly disclosing their annual production of renewables. In addition, we measured the impact of REP on GHG emissions reductions, however, the precise effect of each type of REP on GHG emission reduction has not been covered. Researchers in the future might study the impacts of each type of REP source such as solar, wind and biogas, among others on GHG emissions. Another limitation in this study is that we utilized only the dynamic GMM method in the empirical analysis. Hence, using other models in future research like the threshold model to detect the point of inversion with regard to the relationship between REP and GHG emissions is of paramount importance. Finally, Oil price may palay a significant role in mediating the relationship between REP and GHG Emissions. More specifically, high oil prices could increase affordable capital for renewable energy projects, accelerating the displacement of petroleum and resulting in a stronger reduction in GHG emission. On the flip side, low oil prices could limit budgets for renewable projects and incentivize higher petroleum production to compensate for lost revenue. Therefore, investigating this dynamic relationship is of paramount for future research.

Author contributions
Ridha Ben Saleh, the PhD student has written the main manuscript of this paper. Specifically, the introduction, the data collection and the methodology. Professor Roberto and Professor Nicola has revised each part of this paper and promoted accordingly.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

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